

Senior Flutter Engineer Technical Assessment

Objective

Build a Flutter application that demonstrates a 1:1 real-time video calling experience using the provided backend API and Amazon Chime SDK.

The purpose of this assessment is to evaluate your understanding of Flutter, Android, API integration, state management, and real-time communication. We are not expecting a production-ready application.

Estimated Time: 3-4 hours

Partial completion note -

We understand Amazon Chime SDK integration can involve environment/setup friction. If full two-way video isn't achieved in the time available, submit your best working version and clearly explain the blocker in the README under 'Known limitations.' Partial implementations with clean code and clear reasoning will still be evaluated fairly.

Required Deliverables

Please submit the following:

1. Flutter source code in a Git repository
2. Android APK
3. README with setup instructions
4. Short demo video (maximum 5 minutes)

Functional Requirements

Core Meeting Flow

Implement the following workflow:

- User A creates a meeting
- User B joins the same meeting
- Both users can:
 - View local video
 - View remote video

- Enable/disable microphone
- Enable/disable camera
- Leave the meeting

Meeting States

Display the current meeting status:

- Idle
- Joining
- Connected
- Disconnected

Event Log

Implement a simple event log screen that displays meeting events.

Examples include:

- Meeting started
- Meeting ended
- Participant joined
- Participant left
- Microphone enabled/disabled
- Camera enabled/disabled

A basic scrolling list is sufficient.

Backend Integration

Backend APIs have been provided separately.

Integrate with the provided APIs to:

- Create a meeting
- Join an existing meeting
- Retrieve attendee credentials

Use the provided API documentation and Postman collection. Please integrate only via the provided API endpoints. Do not modify backend/server-side code.

Architecture Requirements

Use one of the following state management approaches:

- Bloc
- Riverpod
- Provider

Organize the project with clear separation between:

- UI
- Business Logic
- API Layer
- Chime Integration Layer

Code Quality

Your submission will be evaluated on:

- Clean and readable code
- Project structure
- State management
- Error handling
- Maintainability

README

Include a short README covering:

- Setup instructions
- State management used
- Brief architecture overview
- Any assumptions made
- Known limitations

Maximum 2 pages.

Demo Video

Provide a screen recording (maximum 5 minutes) demonstrating:

1. Creating a meeting
2. Second device joining the meeting
3. Two-way video communication
4. Microphone toggle

5. Camera toggle
6. Leaving the meeting
7. Event log functioning

Note: Videos that only show source code, GitHub repositories, or project folders without demonstrating the application will not be considered sufficient.

Submission Checklist

Before submitting, ensure you have included:

- GitHub Repository
- Android APK
- README
- Demo Video